

Shivam Rajput

Data Science Intern | Machine Learning | NLP | GenAI and RAG

Dehradun, Uttarakhand | +91 6398994844 | shivamrajput.datascientist@gmail.com

[linkedin.com/in/shivam-rajput-b0407632b](https://www.linkedin.com/in/shivam-rajput-b0407632b) | github.com/shivamrajput-ds | leetcode.com/u/ShivamSynapse

kaggle.com/shivamja | hub.docker.com/u/shivamrajput130

Summary

B.Tech CSE (AI & ML) student targeting Data Science, Machine Learning, NLP, and GenAI/RAG internships, with hands-on projects in large-scale analytics, NLP classification, forecasting, and RAG systems. Built a **15.95M-record** complaint intelligence platform, a hybrid RAG assistant, and an NLP fraud detector using Python, scikit-learn, FastAPI, Streamlit, Docker, and MLflow.

Education

B.Tech - Computer Science & Engineering (AI & ML)

Uttaranchal University, Dehradun

2023 - 2027

CGPA: **8.7 / 10.0**

Technical Skills

Programming & Data: Python, SQL, Pandas, NumPy, Apache Arrow, Parquet, OOP

Machine Learning: scikit-learn, Logistic Regression, Naive Bayes, Random Forest, Gradient Boosting, model evaluation

NLP: Text preprocessing, TF-IDF, n-grams, text classification, topic modeling, model explainability

GenAI/RAG: Vector search, BM25, hybrid retrieval, query expansion, cross-encoder reranking, prompt engineering, LLM integration

Apps & MLOps: FastAPI, REST APIs, Streamlit, Docker, Docker Hub, MLflow, Git, GitHub, testing, CI/CD basics

Databases & Visualization: ChromaDB, Supabase PostgreSQL, Matplotlib, Seaborn, Plotly

Foundations: Statistics, DSA, DBMS, neural network concepts, CNN, RNN, LSTM, GRU, Transformer fundamentals

Experience

Data Science & Machine Learning Project Portfolio

Jun 2026 - Present

Self-Directed Projects

Remote

Customer Complaint Intelligence Platform

[GitHub](#) | [YouTube Demo](#)

Large-Scale Data Science, NLP and Forecasting | Python, Pandas, Apache Arrow, Parquet, Prophet, scikit-learn, Streamlit, Docker

- Processed **15.95M CFPB complaints** from an **8-9 GB CSV** using chunked preprocessing and Parquet storage for memory-efficient analytics.
- Built executive dashboards for complaint trends, product/issue distribution, company behavior, response timeliness, and narrative availability.
- Designed company risk scoring using complaint volume, untimely response rate, and resolution delay, with a minimum-sample floor to reduce misleading rankings.
- Added driver analysis, year-over-year growth tracking, and Prophet forecasting with **3.57% MAPE** on a holdout window.
- Trained NLP routing classifiers with **75.28% Product accuracy** and **62.39% Issue accuracy** using TF-IDF and Logistic Regression.

Enterprise RAG Assistant

[GitHub](#) | [YouTube Demo](#)

GenAI and RAG Document Q&A with Tabular Analytics | FastAPI, Streamlit, ChromaDB, BM25, Docker

- Developed a multi-format RAG assistant for **PDF, CSV, DOCX, Excel, JSON, and TXT** files with document Q&A and structured analytics.
- Implemented hybrid retrieval using vector search, BM25, query expansion, deduplication, and cross-encoder reranking to improve retrieved context relevance.
- Built a Pandas Analytics Engine for CSV and Excel questions involving counts, filters, comparisons, top/bottom records, and numeric conditions.
- Added source visibility, retrieved chunks, feedback capture, and response-time tracking to make RAG answers inspectable instead of black-box LLM responses.

Fake Job Posting Detector

[GitHub](#) | [YouTube Demo](#)

NLP Classification, MLOps and Explainability | Python, scikit-learn, TF-IDF, MLflow, FastAPI, Streamlit, Docker

- Created an NLP fraud detection system on **17,880 job postings** using TF-IDF features and Logistic Regression.
- Compared Logistic Regression, linear support vector classifier, and Naive Bayes; achieved **0.99 ROC-AUC**, **0.88 fraud precision**, **0.84 fraud recall**, and **0.86 fraud F1-score**.
- Integrated MLflow tracking, FastAPI inference, Streamlit dashboard, Docker support, and global/local feature explainability.

Achievements

LeetCode: 511 problems solved | 234 Easy, 242 Medium, 35 Hard | Contest Rating: 1457 | 50 Days Badge 2026